TO: Jury and Judge

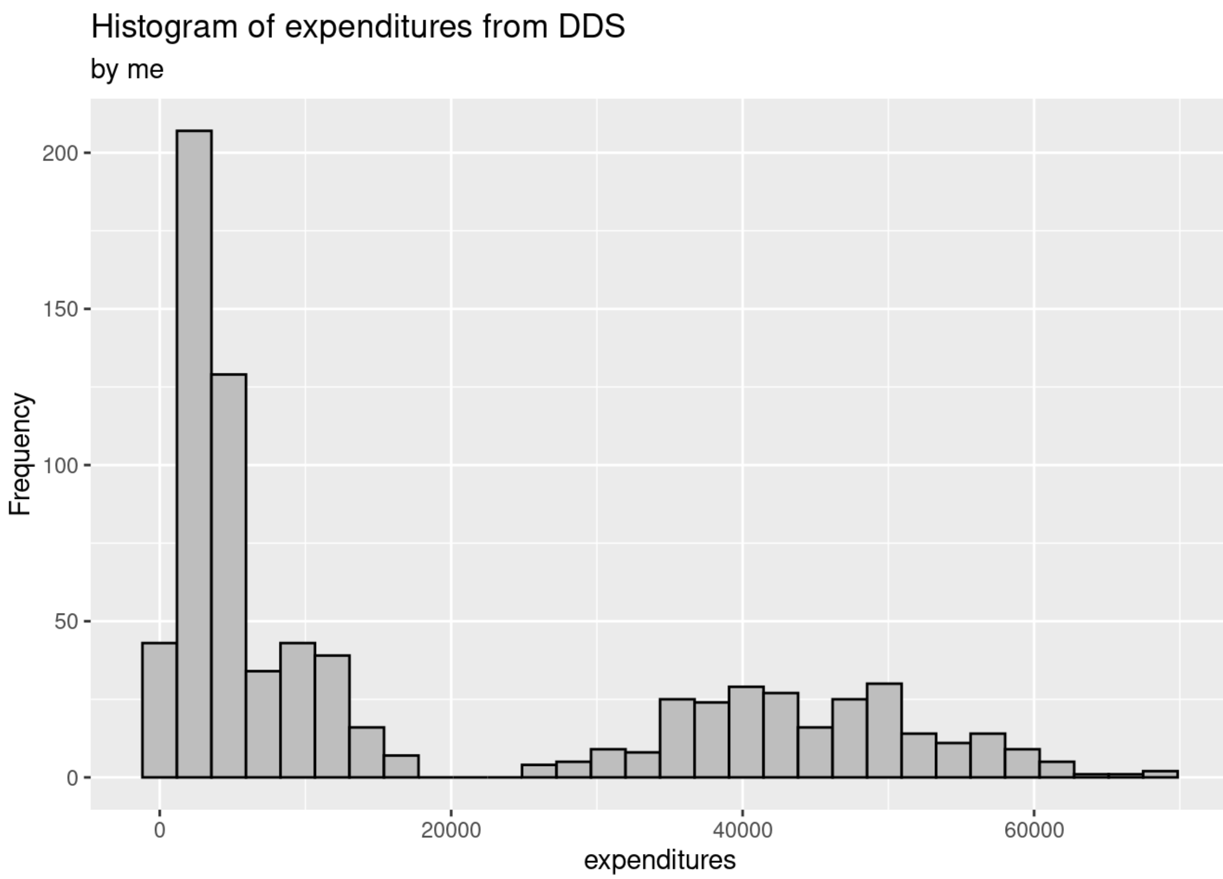
FROM: Shikha Nukala, Senior Analyst

DATE: October 1, 2023

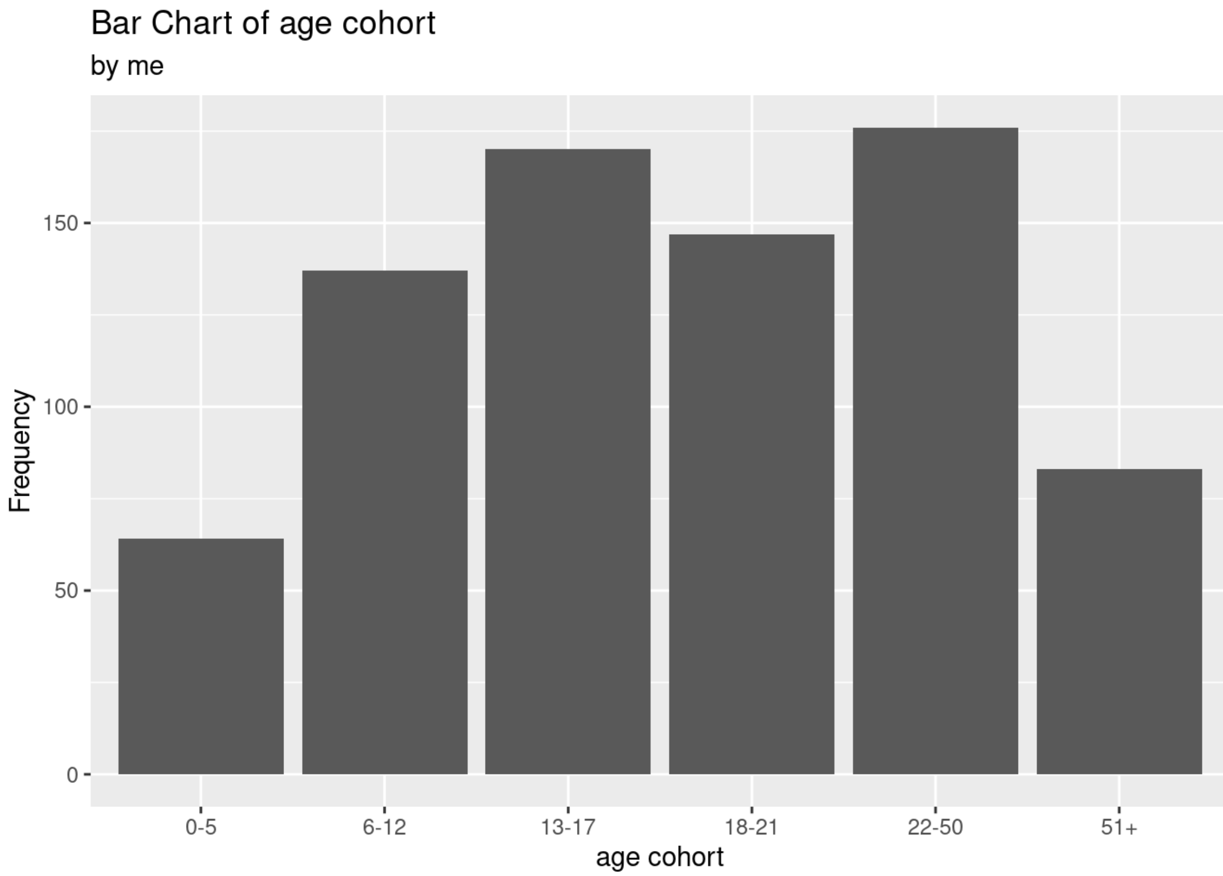
SUBJECT: California DDS Discrimination Case

In this case, we have considered the impact of three related variables that most relate to the case of racial discrimination being brought upon the California Department of Development Services (DDS). To provide a brief overview, we will discuss all variables provided by observational studies that we were given data for: ID of consumer, age cohorts (groups of ages), age of consumer, gender of consumer, expenditures provided by the DDS to each consumer, and the ethnicity of each consumer (Hispanic or White). In this paper, we will identify age as a quantitative discrete variable, age cohort as a categorical ordinal variable, gender as a categorical nominal variable, expenditures as a quantitative discrete variable, ethnicity as a categorical nominal variable, and ID as a **something something** variable. In this case, we identify expenditures as the response variable to ethnicity and age cohort variables which act as the explanatory variables. We also provide a distribution of age cohort against ethnicity, and in terms of our interpretation, we have used age cohort as the explanatory variable to ethnicity as the response variable. Since we have used our statistical analysis to study the expenditures, ethnicity, and age cohort variables, we hypothesize that the age cohort variable may be confounding the effects of the expenditures variable compared with the ethnicity variable in response to the discriminate case basis. We reason that this may be related to more people of older age or younger age receiving more expenditures, potentially offsetting the impact of ethnicity on the expenditures.

For the expenditures variable, we saw from our data analysis that the distribution of expenditures was severely skewed left with most values centered around 6690$ in DDS money received by each consumer. Additionally, it is interesting to note that expenditures are clustered from 20000-60000 for the rest of the data points, showing larger amounts of money given by the DDS for those points (see below for graphical representation).

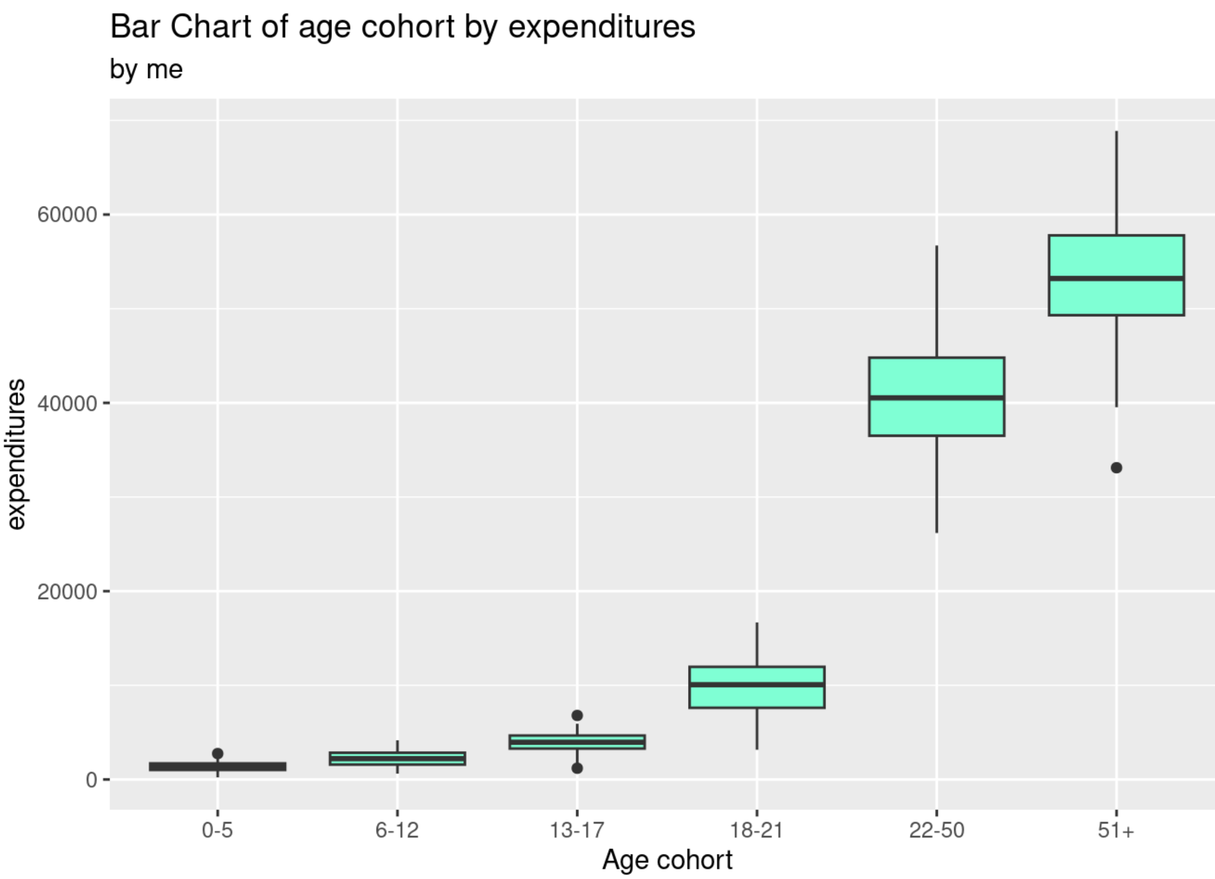


Considering the age cohort variable, we can see that from our data distribution, the most number of people are located in the range of age 22-50 with there being 179 of 777 people from the study noted in the range or a proportion of 0.23. The rest of the sample population is spread out amongst the younger age cohorts, with the 51+ cohort only accounting for 83 of 777 people or a 0.11 proportion (see below for graphical representation).

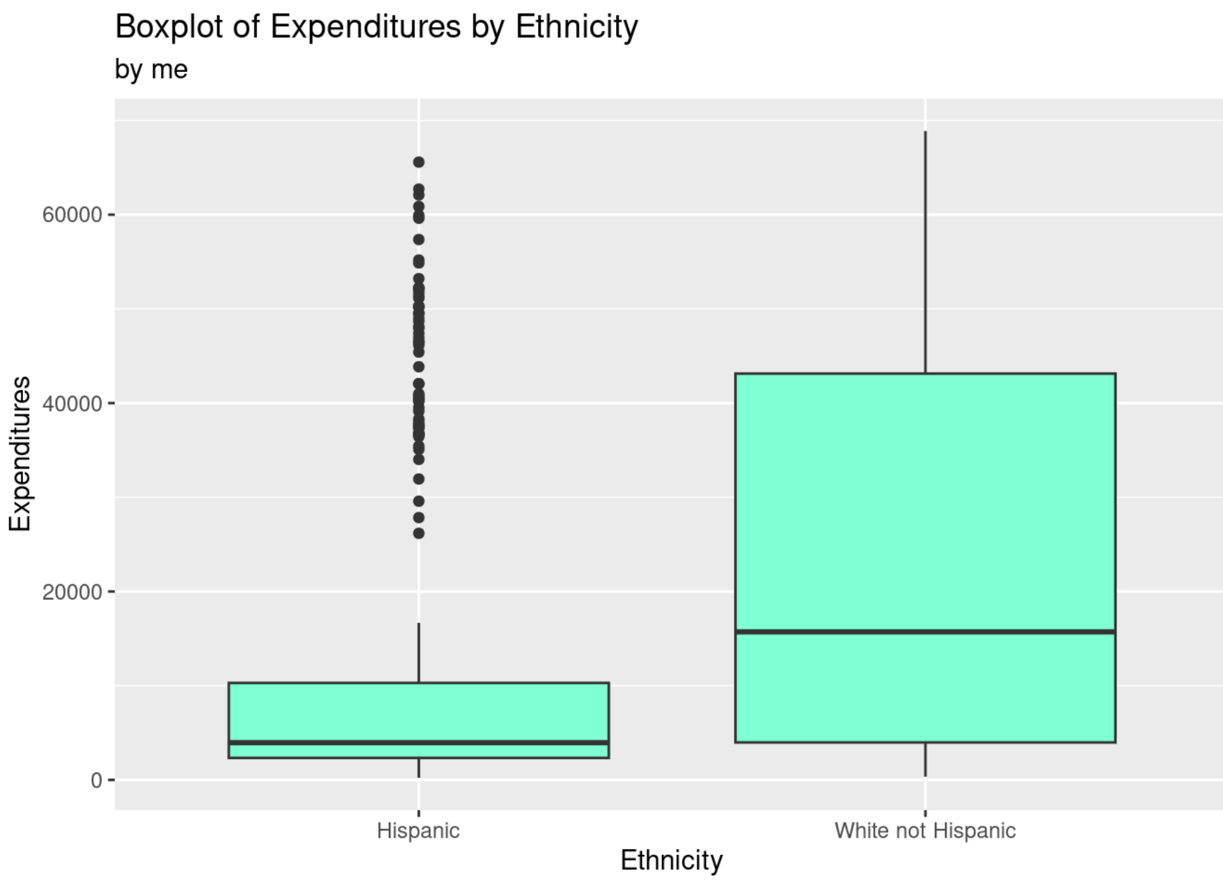


Now that we have reviewed our analyses on two individual variables (we did not conduct a distribution on the ethnicity variable and frequency), we will show our results from conducting two-way variable comparisons (age cohort vs. ethnicity, age cohort vs. expenditures, ethnicity vs. expenditures) and use these findings to provide reasoning for potential racial discrimination.

Starting with the age cohort by expenditures analysis, we can see that there does seem to be an association between the age cohort and expenditures. On average, the higher the age cohort the more the expenditures are based on the medians. We can see that the medians of the 22-50 and 51+ groups are around 40000$ and 54000$ in expenditures respectively compared to the younger age cohort ranges which are all medians below 10000$ in expenditures. The range of the IQRs also increases as age increases, suggesting more expenditures by the DDS on average for higher age cohorts. This makes reasonable sense as older populations either have to provide for children or have to live alone with disabilities, making accessibility harder to get (see distribution below).



Continuing onto the ethnicity by expenditures analysis, we can see that these boxplots seem to confirm the claim that ethnicity is associated with a recipient's expenditure money. For Hispanics, the median expenditures is much lower than the White expenditures given by the DDS (about 1300$: 170000$ for expenditures level respectively). Additionally, the IQRs of each plot indicate that the white ethnicity has a larger range, particularly in higher expenditure numbers, than the Hispanic IQR range indicating that the range for Hispanic expenditures is much lower on average. There are a lot of outliers for Hispanic expenditures, but they are within the range of the white expenditures and the maximum value as per the white distribution. This analysis indicates that Hispanics get less expenditures from the DDS on average compared to the White population who receive expenditures. This association is clear from the distribution of these two variables and provides reasonable evidence to suggest the offense's claim that the DDS has partaken in racial discrimination.We will examine this claim with the third distribution of variables to consider any confounding effect on this association by the age cohort variable (see below for ethnicity and expenditures distribution).



Finally, we can use the first distribution we considered (age cohort against expenditures) and this last distribution for age cohort against ethnicity to check for confounding effects of the age cohort variable on the ethnicity and expenditures variable. From the distribution below, there does seem to be apparent confounding between age cohort and ethnicity. From the graph we can see that age cohort and ethnicity are related with the trends of the higher age cohorts being of more white ethnicity. From the graph, the two age populations from the first distribution with most expenditures per cohort (22-50 and 50+) are comprised of a much larger proportion (nearly 80% White and 20% Hispanic for both age cohorts) than the other age cohorts which on average are comprised of more Hispanic population proportion as seen in the blue color on the bars of the graph. This suggests that higher age cohorts have more White population on average. This association in tandem to the age cohort against expenditures distribution which showed that higher age groups such as 22-50 and 51+ get much more in expenditures overall suggests that higher age cohorts have more expenditures, possibly because of more need because of more severe disabilities amongst older people and being in the position of a caregiver, while high age cohorts are also comprised of more White population, thus showing the association between ethnicity and expenditures as higher for White population on average because the higher age cohorts have more White than Hispanic population. This supports the defense that age cohort does confound the relationship between ethnicity and expenditures, thus providing the misleading assumption from the ethnicity and expenditures association as shown in the variable distribution.

While the ethnicity and expenditures distribution does suggest an association that can lead to a racial discrimination argument that the Hispanic population is provided less expenditures by the DDS as compared to the White population, the age cohort variable confounds the ethnicity and expenditures association because people of higher age tend to get more expenditures than people of lower age, and higher age involves a higher white ethnicity, so both could factor into the expenditures, leading to a defensive support for the DDS that there is not sufficient evidence to claim they engaged in racial discrimination with expenditures.